Tree-glued, cherry-symmetric PDP systems: F4 solving degree against

the caterpillar and full-symmetric arms 2026-11-03 · theory and proposal in `THEORY.md` · generator `gen.sing` (Singular 4.3.2, native) with `instances.txt` · equivalence check `compare.sing` · runner `run.sh` (msolve 0.6.5) · parser `parse.sh` · records `rows.tsv`, `logs/` (sha256 in `logs/SHA256SUMS`) · graph `d_vs_t.svg` (`plot.py`, recorded values only) · PDF `report.pdf`

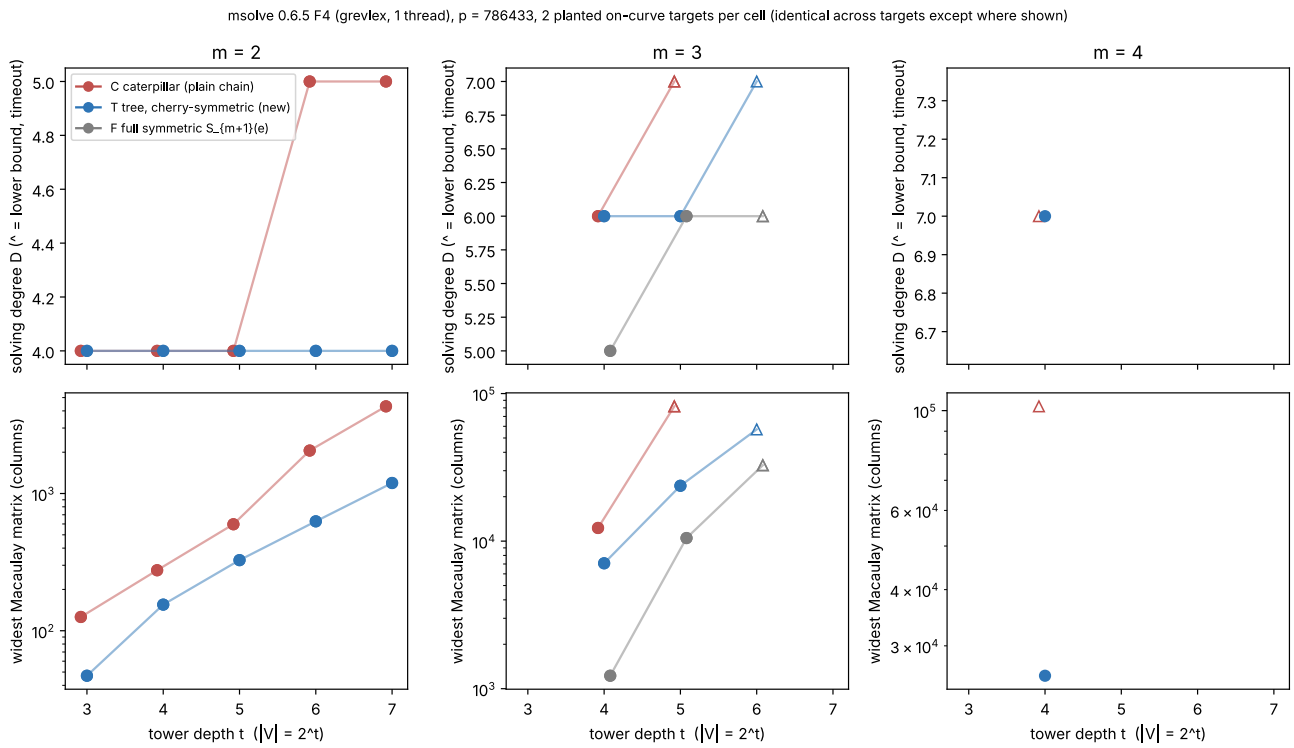
Class: stage diagnostic of the solver axis (AGENTS.md §2). No pipeline is priced, and there is no S value, rho comparison or leaderboard row. It measures one new encoding of the prime-field PDP, motivated by phylogenetic tree models (THEORY.md §3), against the lab's two existing encodings, on identical targets.

Result

The tree-glued cherry-symmetric encoding **T** is new to the lab. Every summand pair ("cherry") is encoded by its symmetric functions (s, q) with a degree-2 remainder tower, and cherries are glued by S_3. All inputs have degree ≤ 4 for every arity m.

- **Against the plain chain C** (same variable count, same targets), T is better in every completed cell:
  - **m = 2:** D stays at **4 for t = 3..7**, while C rises to 5 at t = 6, 7. T's matrices are 1.8–3.6× narrower.
  - **m = 3, t = 5:** T finishes at **D = 6**. C selects degree 7 and is cut off at 600 s with a matrix ≥ 3.4× wider.
  - **m = 4, t = 4:** T finishes at **D = 7** (697 s). C is cut off at 600 s in degree 7 with a matrix ≥ 3.9× wider than T's widest.
  - The pre-registered kill criterion (T's D ≥ C's D in every comparable cell) is **not met**.
- **Against the full-symmetric arm F** (S_{m+1} in e_1..e_m, the lab's symmetric PKM):
  - **m = 3:** F is better: D 5 vs 6 at t = 4, and 2–6× narrower matrices. Both arms reach D = 6 at t = 5.
  - **m = 4, t = 4:** T's D = 7 is **one below** the D = 8 the lab recorded for F (`research/pkm_symmetric_lead1_20261026`, N = 16). But F was 59× faster there (11.8 s vs 697 s) and its matrix was narrower (13,926 vs 25,742 columns). That F row comes from a different engine (`f4_fp`), so only its D is comparable here, and even that loosely.
- **The t-growth is delayed, not removed.** At m = 3, t = 6, T selects degree 7 before the 600 s cut-off. Lead 1's question (is D bounded in t?) stays open for T, as it does for F.

Reading. T dominates the plain chain at every measured cell and gives the first measured cell with a lower solving degree than full symmetrisation (m = 4). It does not beat F's cost at the sizes measured. Its case is asymptotic: F's input degree is 2^{m−1}, while T's is bounded. THEORY.md §3 shows that an L(1/2) cost needs D = O(m) and D nearly flat in t, and only a bounded-input-degree encoding can meet the first condition. Nothing here shows that T meets the second.



Tables (recorded values, `rows.tsv`)

D is the largest selected F4 degree. "≥" marks a run stopped by the 600 s timeout (exit 124); for those runs, D is the degree being reduced when it was cut off, and the width is the widest matrix reached so far. N is the number of unknowns. Both targets gave identical D and widths in every cell where both ran. The two m = 2 C targets differ in sum rows·cols by < 0.1 %.

m	t	N (C, T / F)	D: C	D: T	D: F	widest: C	widest: T	widest: F	quotient dim C / T / F

2	3	6	4	4	= T	126	47		2 / 1
2	4	8	4	4		276	155		2 / 1
2	5	10	4	4		597	327		2 / 1
2	6	12	**5**	**4**		2,056	627		2 / 1
2	7	14	**5**	**4**		4,317	1,193		2 / 1
3	4	13 / 9	6	6	**5**	12,279	7,077	1,224	6 / 3 / 1
3	5	16 / 12	**≥ 7**	**6**	6	≥ 81,920	23,720	10,493	— / 3 / 1
3	6	19 / 15	not run	≥ 7	≥ 6		≥ 57,152	≥ 32,678	
4	4	18	≥ 7	**7** (697 s run)	lab f4_fp: 8	≥ 102,212	25,742	lab: 13,926	— / 6 / —

The quotient dimensions equal the orbit sizes of the planted relation: 2! = 2, 3! = 6 and 4!/4 = 6 copies, against 1 for T at m = 2 and for F. This shows the systems have exactly the planted solution and its relabellings.

Times are contended: up to 5 single-thread jobs shared 4 cores, plus a helper agent. Use them as an ordering only (AGENTS.md: wall time is evidence only from uncontended runs). The 600 s cut-offs are wall-clock, so a contended C or T run could have been cut off earlier than it would have been uncontended. This affects which cells finished, not the D of any completed run. The m = 4 T run with a 2,400 s budget ran with at most two other jobs.

Checks and controls

- **Planted check.** Every system is evaluated inside Singular on a planted decomposition before it is written, and the generator aborts on any nonzero equation. The decomposition uses m distinct points of E(F_p) with x ∈ V, random signs, x_R computed by the group law, and partial sums from the actual points. The curve is $y^2 = x^3 + ax + b$ over F_786433 with (a, b, g) in `instances.txt`.
- **Cross-instrument positive control.** msolve on this generator reproduces the lab's recorded `f4_fp` values from `research/pkm_symmetric_tower_20261026`:
  - plain m = 2: D 4, 5, 5 at N = 10, 12, 14;
  - symmetric m = 3: D 5 at t = 4 and 6 at t = 5;
  - symmetric m = 3, t = 4: widest matrix **1,224 columns, identical**.
- **Generator equivalence.** `gen.sing` was ported from a Python pilot that was kept outside the repository (AGENTS.md: no Python instruments). Its 60 outputs are byte-identical to the pilot's. `compare.sing` checked every polynomial as well: 60 files, 0 mismatches; a deliberately perturbed copy was caught.
- **Parser.** `parse.sh` (awk) agreed with the pilot parser on all 30 logs it was compared on.

What did not run, and why

- **C at m = 3, t = 6–7, and T and C at m = 3, t = 7:** after C was cut off at t = 5 these were not started. The main queue was stopped to free cores for m = 4.
- **m = 4 second target:** C target 1 was killed by the operator at about 300 s, while it was reducing degree 7, to free a core. The partial log is kept as `logs/run600/m4_t4_k1_C.killed.log` and is not used in the table. T target 1 at m = 4 was not run.
- **m = 4, t ≥ 5**, and F at m = 4 with this generator (S_5(e) is not implemented in `gen.sing`). The m = 4 F reference is the lab's recorded row.
- **The S3 resume checkpoint** (`scripts/checkpoint.sh push`) could not upload: the container has no AWS profile and no upload token. `.resume/RESUME.md` was written locally before the runs and holds the pre-registered prediction and kill criterion.
- **Isolab-EKS.** This is a deterministic algebraic diagnostic (degrees, matrix widths, quotient dimensions), and it ran locally. No timing claim is made.

Next measurements

1. **F at m = 4 and 5 against T on one engine** (msolve), so the m = 4 D comparison is not cross-instrument. This is the decisive test of the asymptotic argument: does D_F − D_T grow with m?
2. **T on a larger host at m = 3, t = 6–7 and m = 4, t = 5**, to measure T's t-slope directly.
3. **Separator test of Conjecture B** (THEORY.md §2): in T, replace one internal S_3 by a planted known X and measure how much D drops.
4. **Wreath-complete T:** also quotient the root swap at m = 4 ($S_2 \wr S_2 = D_4$, leaving 3 copies instead of 6).

Two cross-field connections for prime-field index calculus

2026-11-03 · companion to `README.md` (the measurement). **Class: theory note and proposal. No algorithm is priced, and no `S` value or leaderboard row is produced.** The proofs here are sketches unless they are marked "proved". Statements marked "heuristic" are not established.

The lab's prime-field IC work has tried every factor-base symmetry group available to a single instance:

Symmetry	Lab result
S_m on the summands	symmetric Semaev, symmetric PKM, Graeffe and remainder towers
Negation	used throughout
2-torsion translations	FGHR D_m, Montgomery T_0
μ_3 / μ_4 automorphisms	GLV/j=0 orbit bases
Frobenius	binary and extension fields only
Isogeny kernels	generic-group null theorem

The tower presentations tried are the Kummer (μ_{2^t}), Dickson/torus and isogeny (Lattès) towers. This note asks two questions:

1. Why does no factor-base structure lower the elliptic degree?
2. Which symmetry is left that keeps every input degree bounded as the arity m grows?

The connections come from complex dynamics and incidence combinatorics (question 1) and from phylogenetic algebraic statistics (question 2). A grep of both repositories for Elekes, Ritt, "toric fiber product" and phylogenetics finds no hits.

1. The Ritt dichotomy: the three PKM tower families are Ritt's three

families **Ritt's theorem (1923) on permutable maps.** Let f and g be polynomials (or rational maps) of degree ≥ 2 that commute and share no common iterate. Then, up to a common conjugacy, each falls in one of three families:

- power maps $z \mapsto z^n$;
- Chebyshev/Dickson maps T_n;
- Lattès maps, the maps induced on $P^1 = E/\pm 1$ by an endomorphism of an elliptic curve E.

Observation (proved; elementary). The lab's three tower families are exactly these three families, each attached to a one-dimensional algebraic group:

lab tower	self-map	algebraic group it is an endomorphism of
Kummer, V = g·μ_{2^t}	$w \mapsto w^2$	G_m (the smooth locus of the **nodal** cubic)
Dickson / norm-one torus	$x \mapsto x^2 - 2$	G_m/±1, i.e. the Kummer line of the nodal cubic
isogeny / Lattès tower	$x \mapsto x([2]P)$ or $x(\varphi(P))$	E/±1 itself

Only the third family is compatible with the elliptic group law. Each Kummer or Dickson tower is a doubling tower for a different group: the multiplicative group of the nodal degeneration $y^2 = x^2(x+c)$ of E. The summation polynomials of that degenerate group are linear in log coordinates and interact with the tower. The summation polynomials of E do not.

Lemma A (proved; rigidity). Let $\Phi : E \longrightarrow E$ be a morphism of varieties such that $\Phi(P)+\Phi(Q)+\Phi(R) = O$ whenever $P+Q+R = O$. Then $\Phi = \varphi$
1. T with $\varphi \in \mathrm{End}(E)$ and $3T = O$.

Proof: every morphism $E \longrightarrow E$ is a translate of a homomorphism. Then $3T = \Phi(O)+\Phi(O)+\Phi(O) = O$. ∎

So a tower that preserves the summation variety level by level is a Lattès tower. Its factor base F = Φ_t^{-1}(S) is a union of cosets of ker Φ_t. A relation over F is then a relation over E/ker Φ_t on the image of S, which is the lab's generic-group null theorem (RESEARCH_ISOGENY_FIBERS_ECDLP.md Thm 15.1). A tower that does not preserve it is, by Ritt, a power or Chebyshev tower, which belongs to the nodal group.

Dichotomy (heuristic as stated; Lemma A is its rigorous half). A factor-base tower is either:

- compatible with the group law, in which case the base degenerates to IC on an isogenous curve; or
- incompatible, in which case it is a tower for the nodal degeneration and gives no elliptic cancellation in S_{m+1}.

This predicts what the lab measured:

- the solving degree D of every Kummer/Dickson tower rises with the depth t (staircase 4, 5, 5, 6, 6 at m = 3);

- relation counts on structured bases equal those of random bases (lever-2 null, 5,760 instances).

2. Elekes–Szabó: the quantitative version

Elekes–Szabó (2012), and the finite-field variants of Tao and of Bays–Breuillard. Let F(x,y,z) be an irreducible surface. Either $|Z(F) \cap A\times B\times C| \le O(n^{2-\eta})$ for all n-sets A, B, C, or F is locally the graph of addition in a one-dimensional algebraic group: G_a, G_m or an elliptic curve. In the special case, the large intersections come from approximate subgroups (progressions) of that group.

The S_3 surface is special: it *is* the elliptic group law. So a factor base A with many more relations than random must be close to a coset progression of E(F_p). Writing down such a set needs discrete logarithms. A Kummer base is a progression in G_m, the wrong group. For a set that is multiplicatively but not elliptically structured, the theorem gives an upper bound on the relation count over the random expectation; it does not give an enhancement. That is the lever-2 null. The finite-field statements hold only for |A| below a power of p, so this is the heuristic reading at IC sizes, not a proof.

What it says about the degree (heuristic). In any tree-shaped PDP system, the separators are the partial-sum abscissas X = x(P_i + P_j). Their elimination ideal is the pair-sum set $\Sigma_2(V) = \{x(P+P') : P, P' \in F\}$. By Elekes–Szabó this set is "generic": it has size about $|V|^2$ and no low-complexity presentation. F4 has to reach this degree, or the analogous degree for larger partial sums. **Conjecture B:** the t-growth of D comes from the separators, not from the membership towers. Every lab measurement agrees with it, but none tests it directly. The direct test is to plant a separator constraint and compare D.

3. Phylogenetic gluing: the symmetry that is left

Algebraic statistics. In a group-based phylogenetic model on a tree, each internal vertex carries the group sum of its children's states. Its ideal is glued from the star ideals of the tree's vertices along the internal edges. These are Sturmfels–Sullivant (2005) and Sullivant's toric fiber products (2007). For codimension-zero toric fiber products, a Gröbner basis of the glued ideal is assembled from those of the pieces, in degree at most the pieces' maximum. The model's automorphism group is the automorphism group of the tree; for a balanced rooted binary tree with m = 2^k leaves it is the iterated wreath product $S_2 \wr \dots \wr S_2$, which is the Sylow 2-subgroup of S_m.

The ECDLP relation is a group-based tree model. A decomposition $R = P_1 + \dots + P_m$ is a rooted tree: the leaves are factor-base points, the internal vertices are partial sums, and each vertex carries the elliptic group law of its children, which is S_3. The lab's two families are two extremes:

- **plain chain (caterpillar).** The tree is a path, its automorphism group is trivial, and its inputs have degree ≤ 4 for every m. Each relation is seen m! times.
- **full symmetric (S_{m+1}(e)).** It quotients all of S_m but has one vertex of degree 2^{m−1}, so its input degree, and with it D, is exponential in m. The lab measured D = 8 at m = 4.

The tree-glued cherry-symmetric system (this proposal). Pair the summands into cherries of a balanced tree. Encode each cherry (P_{2c−1}, P_{2c}) by the symmetric pair (s_c, q_c) = (x+x', xx'). Its membership is the degree-2 remainder tower $T^{2^t} \equiv 1 \bmod (T^2 - s_cT + q_c)$. Its vertex is $S_3(s_c, q_c, X_c)$, the symmetrized S_3 of degree 4. Internal vertices glue the X's by S_3.

- This quotients $(S_2)^{m/2}$ of the wreath symmetry, i.e. 2^{m/2} copies.
- It keeps **every input at total degree ≤ 4 for every m**.
- It has the same variable count as the caterpillar.

As far as the lab's records show, it is the only one of the three encodings that both removes a growing part of the symmetry and has bounded input degree.

Why bounded input degree is the condition that matters for a subexponential cost. Take $|F| \approx p^{1/m}$, $N \approx m\cdot t + O(m)$ unknowns and $t = \log_2 p / m$. One PDP costs about $\binom{N+D}{D}^\omega$, there are about m! targets per relation, and the relations number about |F|. The total is $L_p(1/2)$-subexponential only if $\log \binom{N+D}{D} = O(\sqrt{\log p \cdot \log\log p})$ at $m \approx \sqrt{\log p / \log\log p}$. Since $N \approx \log p$, that needs:

1. **D = O(m) up to log factors, which rules out any encoding with a degree-2^{m−1} input** and
2. **D nearly independent of the depth t**, because $t \gg m$ at the optimum.

Condition 1 rules out the full symmetric arm asymptotically, whatever it gains at small m. Condition 2 is what the measurement tests: the *t-slope* of D at fixed m. The full gluing theorem would give D bounded by the cherry D. It is not expected to hold, because the S_3 gluing is not toric and Conjecture B predicts separator growth. The measurement asks whether the tree arm's t-slope is below the caterpillar's.

Kill criterion (pre-registered in `.resume/RESUME.md` before the solver runs). The tree's D is at least the caterpillar's in every comparable cell.